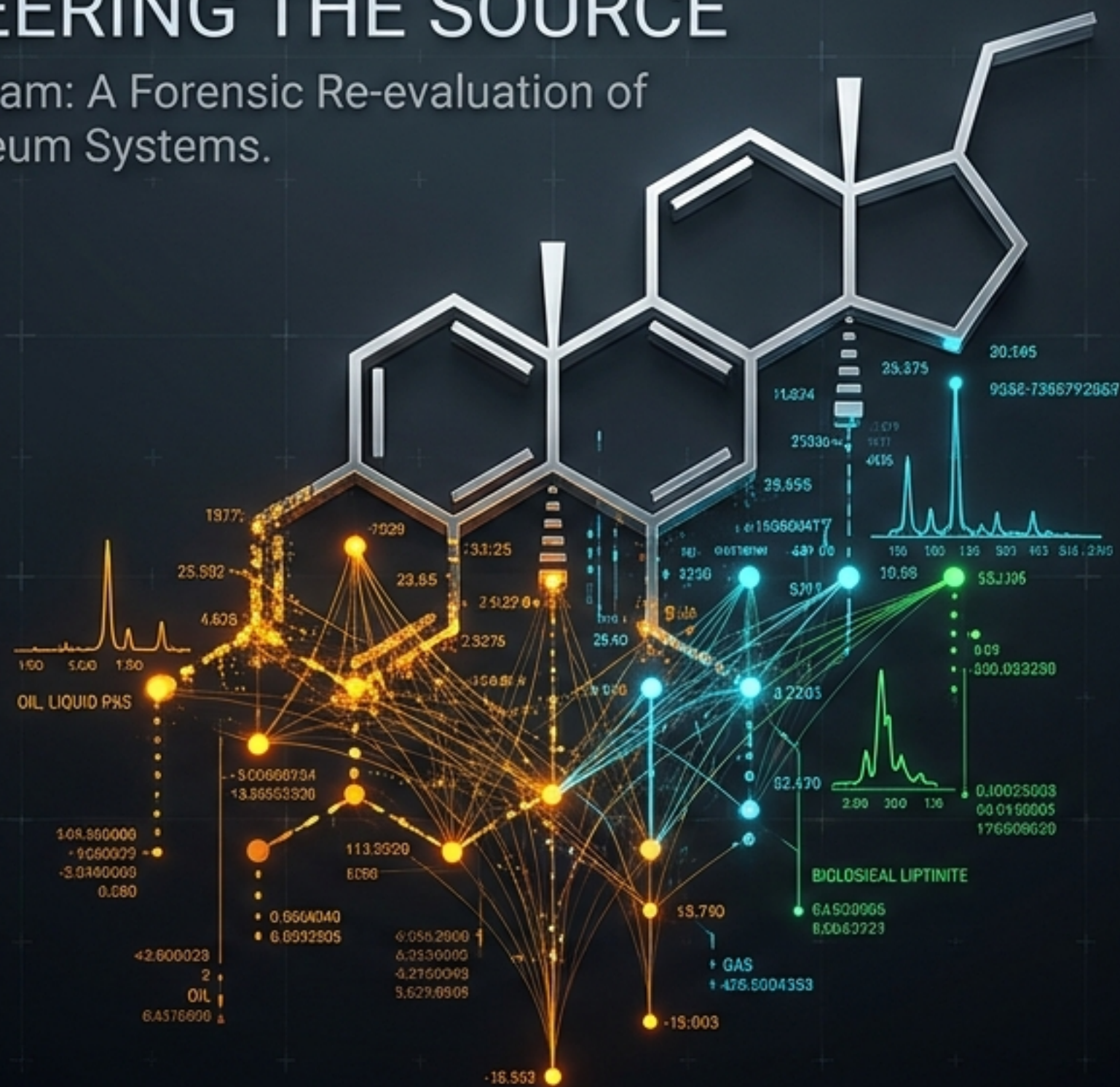


REVERSE-ENGINEERING THE SOURCE

Beyond the Van Krevelen Diagram: A Forensic Re-evaluation of Kerogen, Macerals, and Petroleum Systems.



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Hydrocarbons are chemical memories.

Traditional analysis moves from
deposition to maturation to generation.

**Forensic petroleum system analysis
reverses the flow.**

We analyze the **physical phase** and
chemical composition of these fluids to work
deconstructing the **basin's thermal history**
and **original biological architecture**.

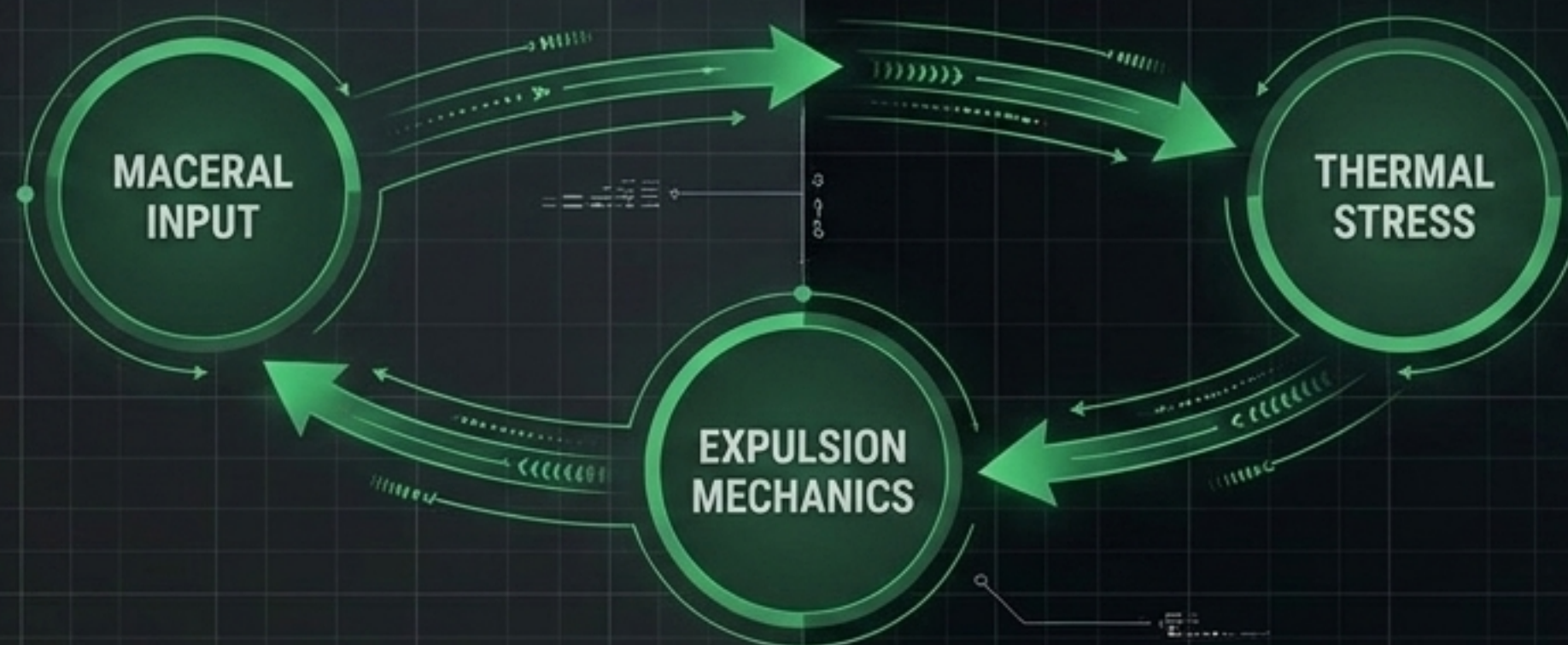


The Ultimate Variable

[KEROGEN TYPE CONTROLS EVERYTHING.]

From **oil-prone Type I & II** (oil) to **gas-prone Type III** (cyan) and inert **Type IV** — the maceral composition tells us what the source rock will generate and when.

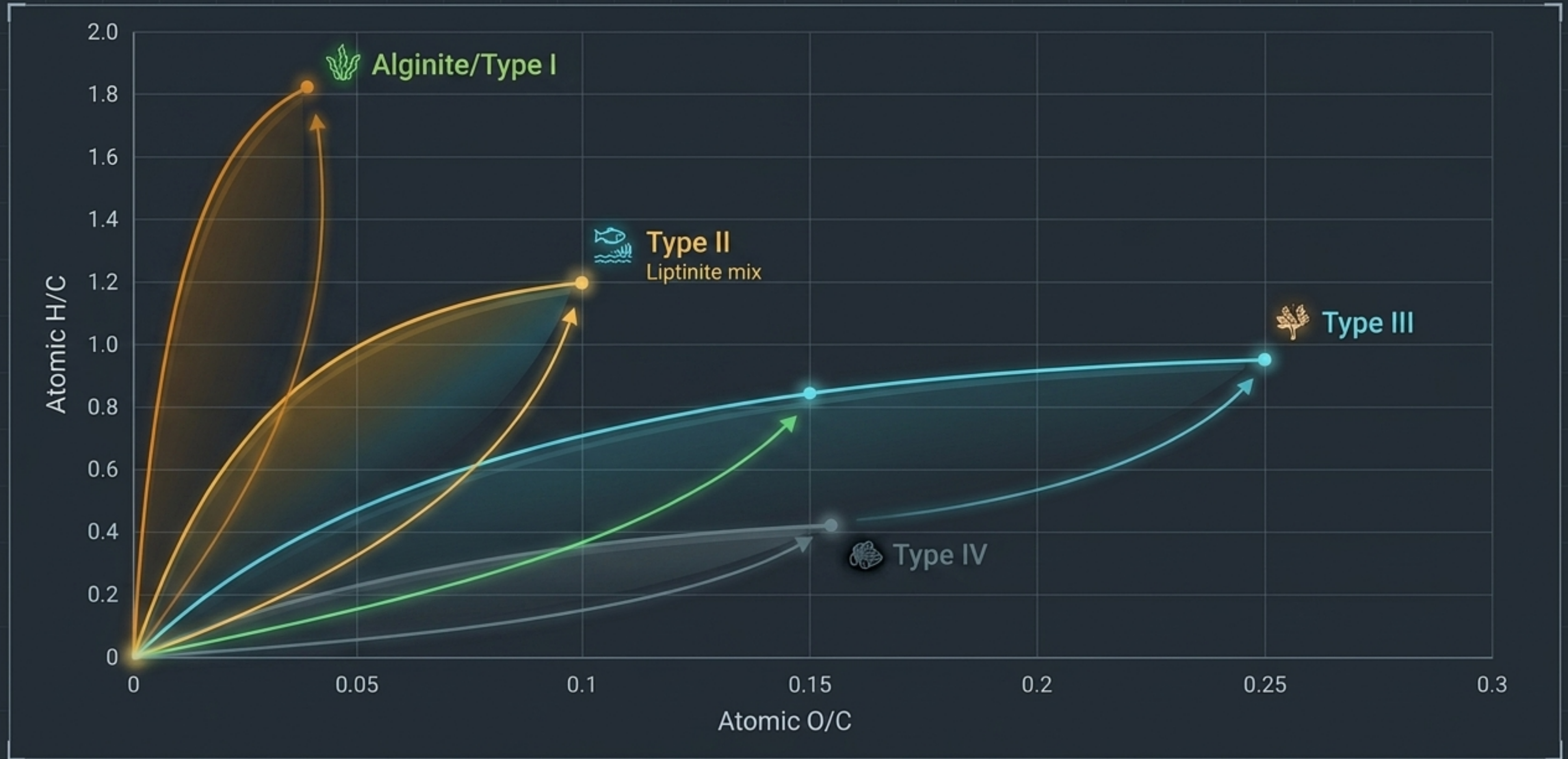
But static classification creates blind spots. Understanding the link between macerals and kerogen is fundamental, but treating 'control' as a static rule ignores the dynamic chemistry of mixed assemblages and systemic basin evolution.



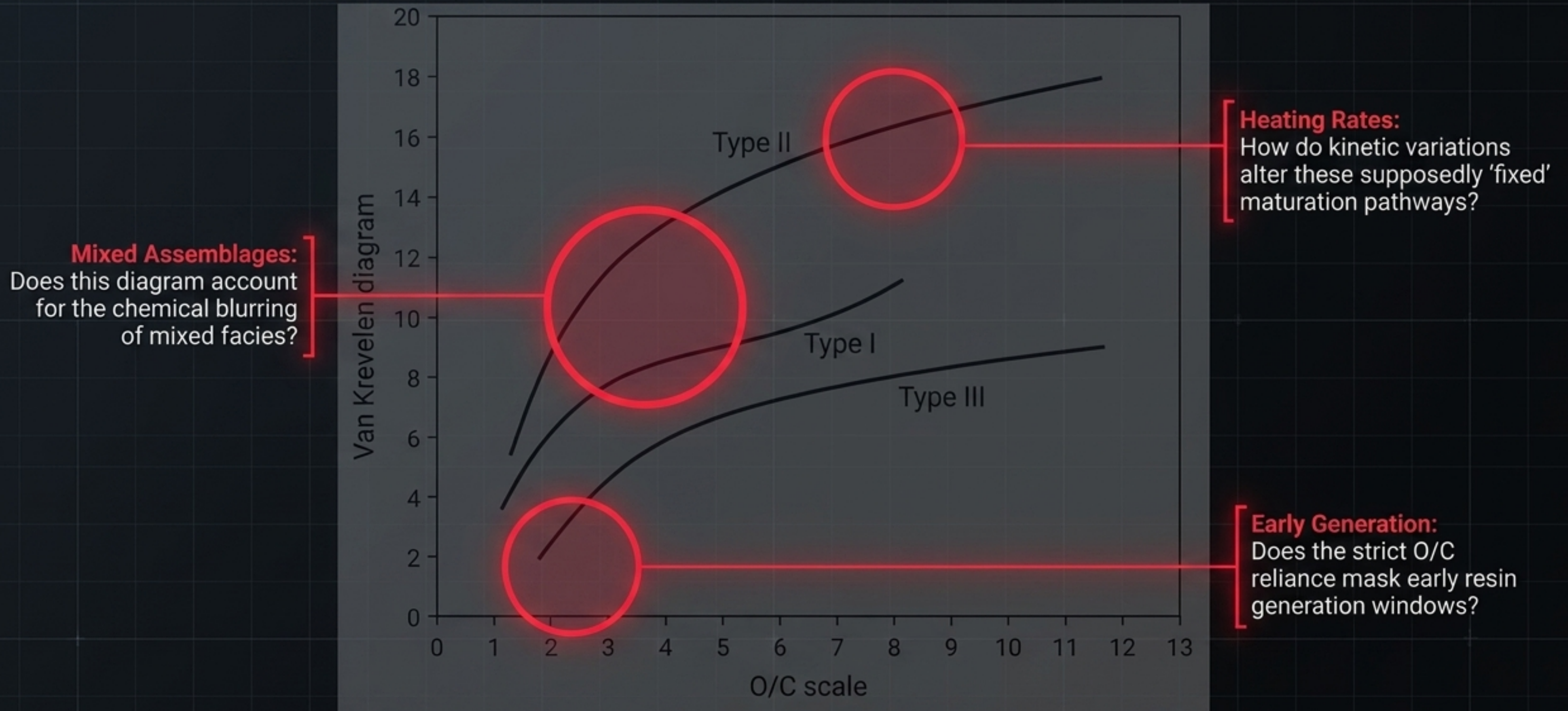
The Standard Architecture of Organic Matter



The Van Krevelen Cartography

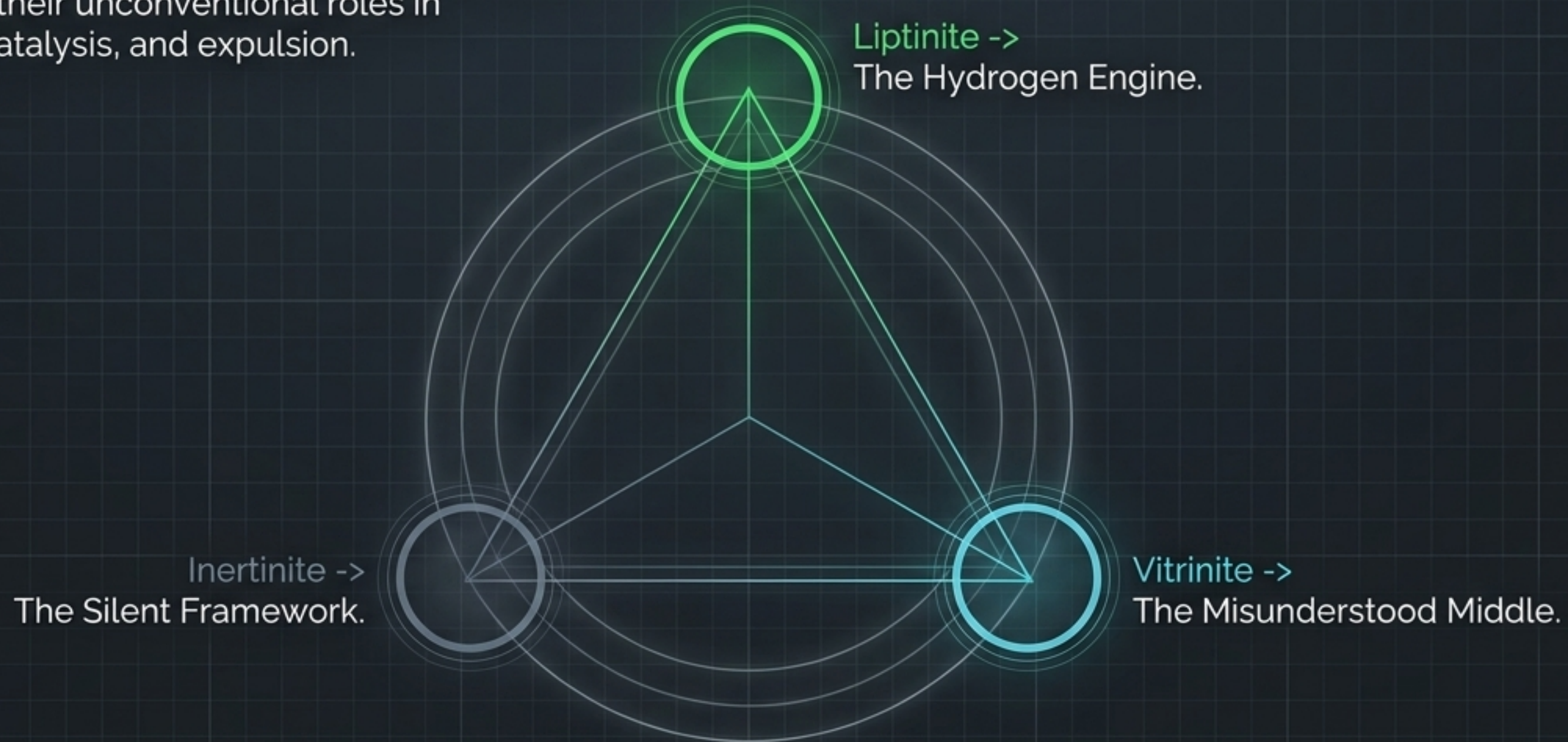


Questioning the Cartography



The Maceral Triad: Beyond Biological Definitions

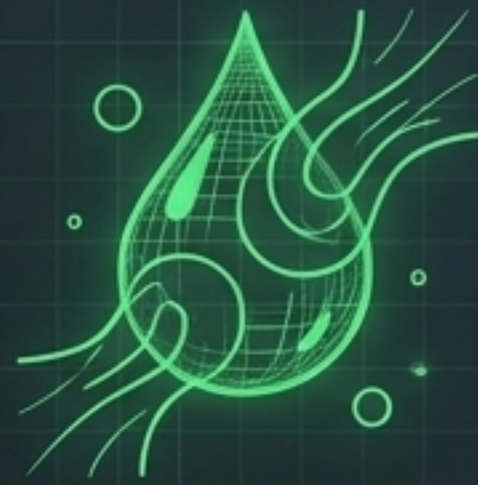
To truly evaluate source rocks, we must abandon simple input-output memorization. We must analyze these macerals for their unconventional roles in generation, catalysis, and expulsion.



Liptinite: The Hydrogen Engine



Alginite



Resinite



Cutinite



Sporinite

Unconventional Angle: Move beyond "Type I/II generates oil." **Resinite** can act as an **early-generation catalyst**, expelling **liquids** well before the standard oil window. **Alginite** acts as a **massive hydrogen donor**. These are **active chemical engines**, not just passive inputs.

Vitrinite: The Misunderstood Middle

The Orthodox View

Collinite & Telioite
(Wood/Bark origins)



Strictly Type III



Strictly Gas-prone.

The Forensic View



Reading between the lines: Hydrogen-rich vitrinites (often deposited in specific humic conditions) possess hidden **liquid generation potential**. Treating Vitrinite as 'gas-only' results in severely **underestimated liquid reserves** in mixed terrigenous basins.

Inertinite: The Structural Framework

THE ORTHODOX VIEW

Fusinite, Semifusinite, Funginite, Micrinite.

Classified as Type IV.

Standard definition: Inert, oxidized, zero generation potential.

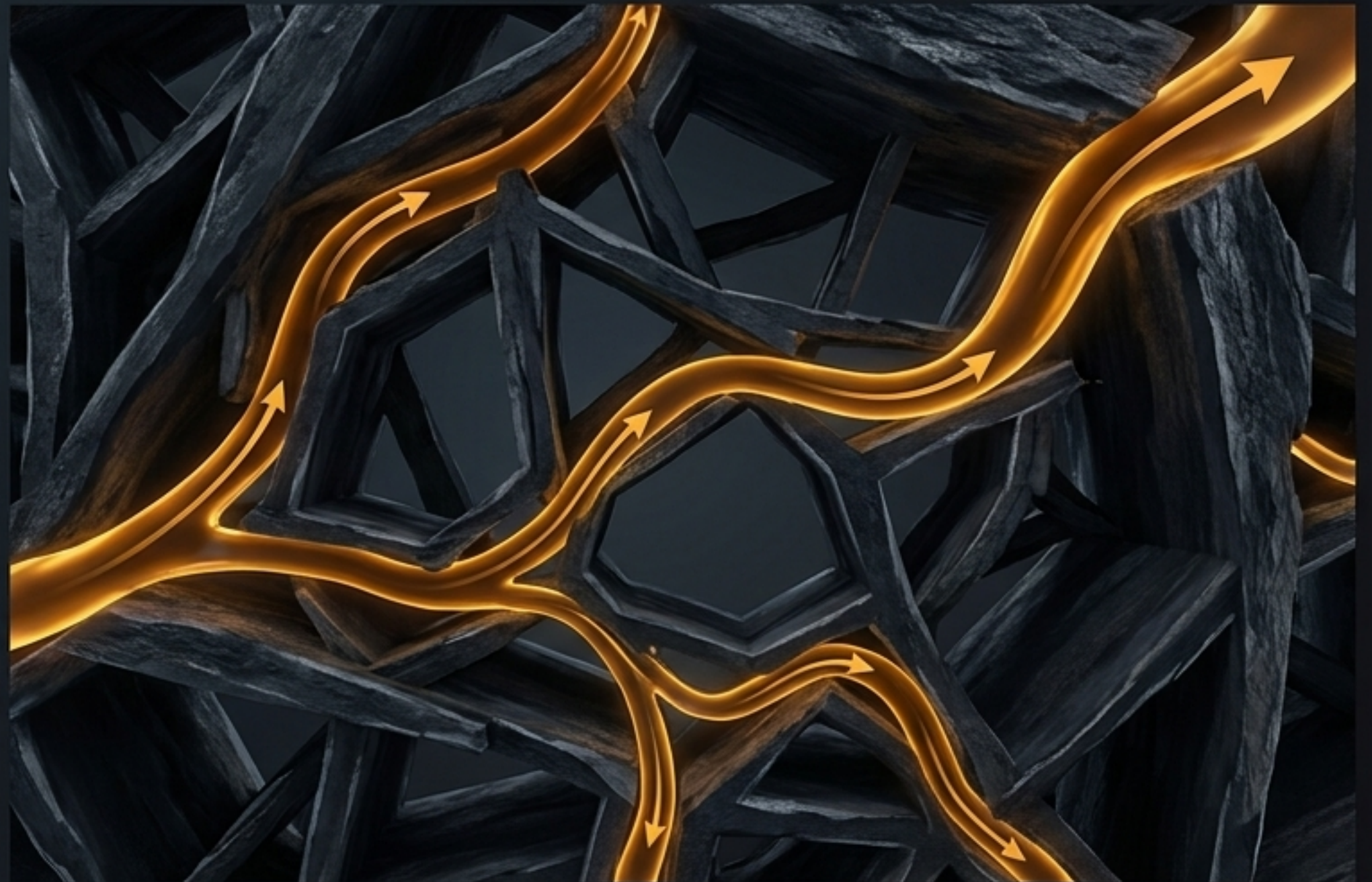
REVERSE THINKING

While chemically inert, it is mechanically vital. Inertinite provides a **rigid micro-porosity framework**.

It prevents total compaction of the source rock, acting as the **microscopic highway** that allows generated hydrocarbons to successfully migrate and expel.

It is not dead weight; it is the **infrastructure**.

THE STRUCTURAL FRAMEWORK ILLUSTRATION

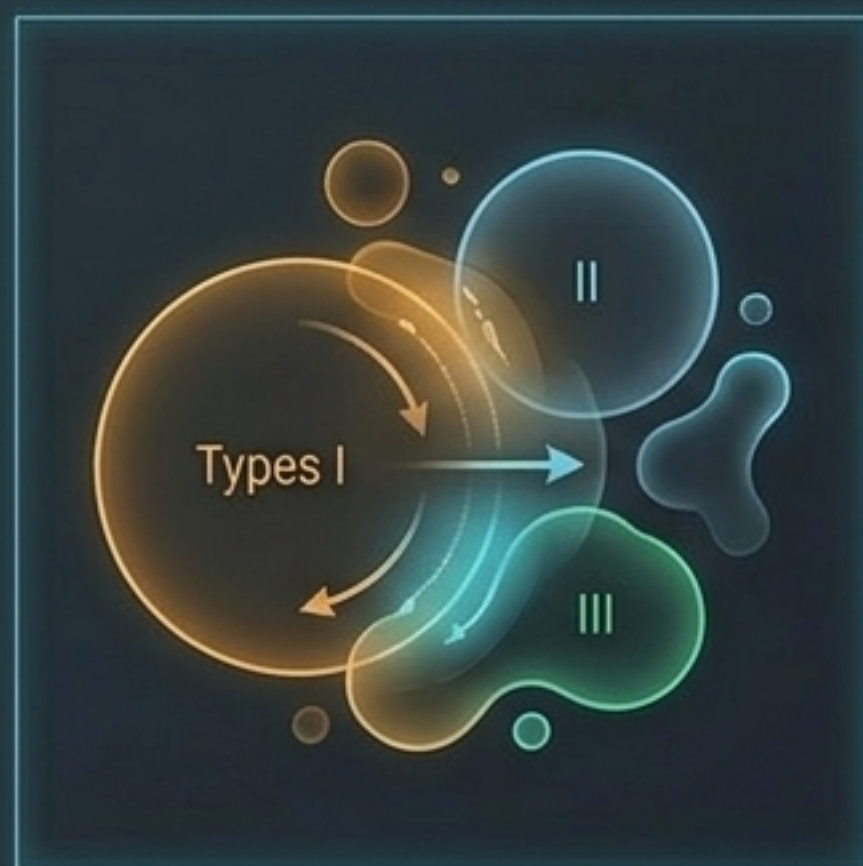


The Dynamic Maceral Diagnostic Matrix

Maceral Group	Standard Output	Unconventional System Role	Van Krevelen Blindspot
Liptinite (Alginite, Resinite)	Oil-prone	Early thermal catalysts / H-Donors	H/C ratio masks early resin generation windows
Vitrinite (Collinite, Telioite)	Gas-prone	Liquid generation in specific humic facies	Overlap with Type II in mixed facies
Inertinite (Fusinite, Micrinite)	Inert	Expulsion infrastructure (micro-porosity)	Treated as chemical diluent; ignores mechanical impact

Redefining Petroleum System Analysis

Step 1: Discard Static Boundaries



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Stop using hard lines between Types I, II, and III. Model transitional kinetics based on exact maceral sub-types.

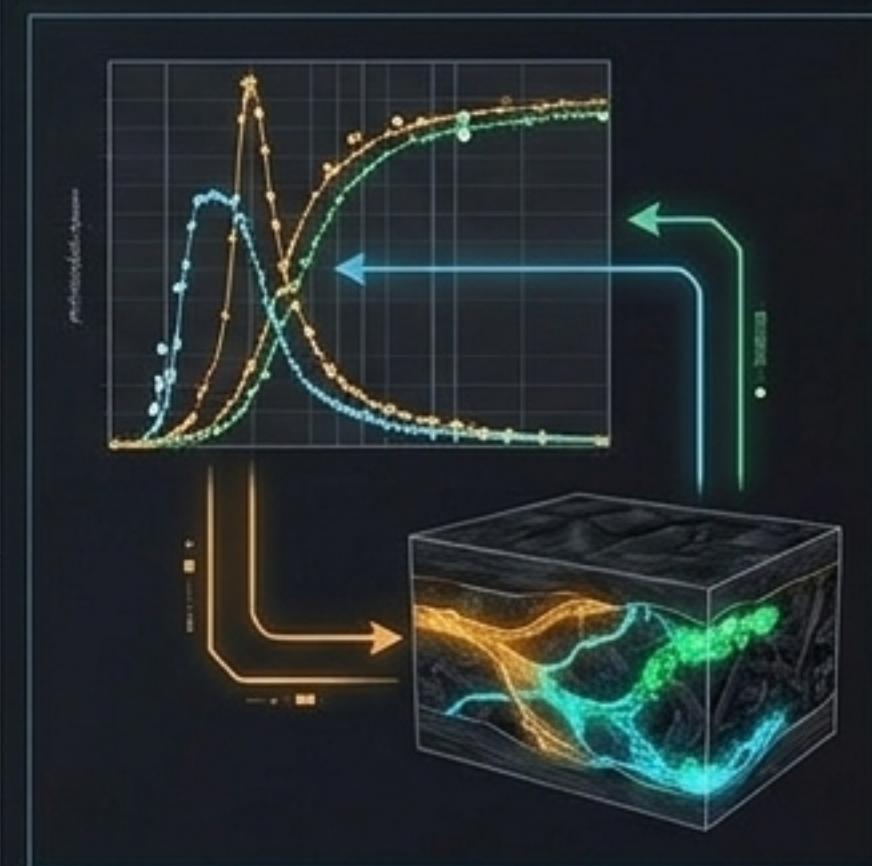
Step 2: Model the Mechanics



Step 2: Model the Mechanics

Incorporate the structural volume of Inertinite into expulsion efficiency models, not just TOC calculations.

Step 3: Reverse-Engineer the Facies



Step 3: Reverse-Engineer the Facies

Use produced fluid signatures to backtrack and confirm the presence of early-generating resins or hydrogen-rich vitrinites.

From Fluid to Foundation

Kerogen type controls everything—but only if you understand the dynamic, living system that built it. True source rock evaluation requires seeing the ecosystem within the oil.

